

Challenge (Habanero)

- Q1.** A park ranger is tracking the population of deer in a forest. If the total number of deer is 45, what is the remainder when this number is divided by 9?
(A) 0
(B) 1
(C) 3
(D) 6
(E) 9
- Q2.** A hiker is planning a trip that requires 540 miles of driving. If the car's fuel consumption is 20 miles per gallon, how many gallons of fuel will be consumed, and what is the remainder if the total gallons are divided by 5?
(A) 26 R 2
(B) 27 R 0
(C) 27 R 2
(D) 27 R 3
(E) 28 R 1
- Q3.** What is the remainder when 98 is divided by 7?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q4.** A tour guide is arranging 48 people into groups for a hike. If each group can have 8 people, and 4 people are left out, what percentage of the total number of people is left out?
(A) 5
(B) 6
(C) 7
(D) 8
- (E) 9
- Q5.** What is the remainder when 117 is divided by 10?
(A) 5
(B) 6
(C) 7
(D) 2
(E) 9
- Q6.** A group of friends want to share some candy equally. If they have 63 pieces of candy and there are 9 friends, how many pieces of candy will each friend get, and what is the remainder?
(A) 6 R 6
(B) 7 R 0
(C) 6 R 9
(D) 7 R 1
(E) 7 R 0
- Q7.** What is the remainder when 75 is divided by 5?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q8.** A camper has 48 feet of rope to set up a tent. If the rope is divided into pieces of 8 feet each, how many pieces can be made, and what is the remainder?
(A) 5 R 4
(B) 5 R 6
(C) 6 R 0
(D) 6 R 1
(E) 6 R 4

Open Ended Questions (FRQ)

- Q9.** A conservationist is studying the population of rabbits in a forest. The initial population of rabbits is 240. The population increases by 15 each year. Determine the number of years required for the population to reach 300 rabbits. Show all steps and calculations.
- Q10.** A traveler is planning a road trip that requires 1200 miles of driving. If the car's fuel consumption is 25 miles per gallon, and the car has a gas tank capacity of 20 gallons, determine how many times the traveler will need to refill the gas tank, assuming the tank is empty at the beginning of the trip. Show all steps and calculations.

Chef's Tips

- Provide specific examples to illustrate each concept
- Encourage students to use real-world scenarios for word problems
- Emphasize the importance of showing all steps and calculations for free-response questions